MODULE 4 ALIMENTARY CANAL

The Alimentary Canal.

THE DIGESTIVE SYSTEM, THE ALIMENTARY CANAL

Covers the wall of the digestive tube, the peritoneum, and the regions of the canal from the mouth to the anus, with the named features of the stomach and the intestines. The focus is on the structures and the job each one does.

BY THE END

1. Distinguish the alimentary canal from the accessory organs and name the four layers of its wall.
2. Trace the pathway of food from the mouth to the anus.
3. Identify the named features of the stomach, small intestine, and large intestine.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The digestive system

drsrennie-stack.github.io/new-build-bio4-solano/gi-concept-videos.html

The Alimentary Canal, an Overview

The digestive system has two groups of organs. The alimentary canal is the continuous tube that food travels through; the accessory organs sit alongside it and empty their secretions in. Covers the canal.

THE TWO GROUPS OF DIGESTIVE ORGANS

Structure	What it is
Digestive system	The organs that take in food and eliminate waste; the alimentary canal plus the accessory organs
Alimentary canal	The gastrointestinal tract; a continuous muscular tube from the mouth to the anus
Organs of the canal	The mouth, most of the pharynx, the esophagus, the stomach, the small intestine, and the large intestine
Accessory organs	The teeth, tongue, salivary glands, liver, gallbladder, and pancreas; covered on their own sheet
Lumen	The open interior space of the tube, continuous with the outside world at the mouth and anus
Length	About 16 to 23 feet in a living person, longer once muscle tone is lost

The Wall of the Alimentary Canal

From the lower esophagus to the anal canal, the wall of the tube is built from the same four layers, deep to superficial. Compare them.

THE FOUR LAYERS OF THE WALL

Layer	What it contains
Mucosa	The inner lining; an epithelium in contact with food, the lamina propria of areolar connective tissue beneath it, and a thin muscularis mucosae
Submucosa	Areolar connective tissue with blood and lymphatic vessels, and the submucosal nerve plexus
Muscularis	Usually an inner circular and an outer longitudinal layer of smooth muscle, with the myenteric nerve plexus between them
Serosa	The outermost layer; a serous membrane, also called the visceral peritoneum, replaced by a fibrous adventitia on the esophagus

Two nerve networks in the wall make up the enteric nervous system, the nerve supply of the gut.

THE ENTERIC NERVOUS SYSTEM

Structure	What it is
Submucosal plexus	The nerve network in the submucosa that controls the secretions of the gut
Myenteric plexus	The nerve network between the muscle layers that controls the movement of the gut

KEY POINT

Four layers everywhere, with two exceptions worth holding. The esophagus wears a fibrous adventitia where the rest of the tube has a serosa, and the stomach carries a third layer of muscle where the rest carries two.

The Peritoneum

The peritoneum is the large serous membrane of the abdomen. Its folds anchor the digestive organs and carry their vessels and nerves.

THE LAYERS OF THE PERITONEUM

Structure	What it is
Parietal peritoneum	The layer that lines the wall of the abdominal cavity
Visceral peritoneum	The layer that covers the organs; it is the serosa of those organs
Peritoneal cavity	The slim, fluid-filled space between the parietal and visceral layers
Retroperitoneal organs	Organs behind the peritoneum, not within its cavity; the kidneys, pancreas, duodenum, and parts of the colon

Five large folds of peritoneum weave among the organs. Compare what each one connects.

THE FIVE FOLDS OF PERITONEUM

Fold	What it connects
Greater omentum	The largest fold; a fatty apron hanging from the stomach over the intestines
Lesser omentum	Connects the stomach and duodenum to the liver, and carries vessels to the liver
Falciform ligament	Attaches the liver to the anterior abdominal wall and the diaphragm
Mesentery	A fan-shaped fold binding the jejunum and ileum to the posterior abdominal wall
Mesocolon	Binds the transverse and sigmoid colon to the posterior abdominal wall

The Mouth, Pharynx, and Esophagus

Food enters at the mouth, is swallowed through the pharynx, and travels the esophagus to the stomach.

FROM THE ORAL CAVITY TO THE STOMACH

Structure	What it is
Oral cavity	The mouth; bounded by the cheeks, the lips, the palate, and the tongue
Hard palate	The bony anterior roof of the mouth, formed by the maxillae and palatine bones
Soft palate	The muscular posterior roof of the mouth
Uvula	The finger-like projection of the soft palate that closes off the nasopharynx during swallowing
Fauces	The opening from the mouth into the oropharynx
Palatine and lingual tonsils	Lymphatic tissue between the palatal arches and at the base of the tongue
Pharynx	The funnel-shaped passage with three regions: the nasopharynx, oropharynx, and laryngopharynx
Esophagus	The collapsible muscular tube, about 10 inches long, that carries food from the laryngopharynx to the stomach
Esophageal hiatus	The opening in the diaphragm that the esophagus passes through
Upper and lower esophageal sphincters	The rings of muscle that guard the top and bottom of the esophagus; the lower is also called the cardiac sphincter

The Stomach

The stomach is a J-shaped widening of the tube below the diaphragm. It is a mixing chamber and a holding reservoir.

THE REGIONS AND NAMED FEATURES OF THE STOMACH

Structure	What it is
Cardia	The region that surrounds the opening from the esophagus
Fundus	The rounded portion that bulges above and to the left of the cardia

Structure	What it is
Body	The large central portion of the stomach
Pyloric part	The region that connects the body to the duodenum; it has the pyloric antrum, pyloric canal, and pylorus
Pyloric sphincter	The ring of smooth muscle between the pylorus and the duodenum
Lesser curvature	The concave superior border of the stomach
Greater curvature	The convex inferior border of the stomach
Rugae	The folds of the inner stomach lining that flatten as the stomach fills
Gastric pits and glands	The pits in the stomach lining that lead to the gastric glands, the secreting glands of the stomach
Muscularis of the stomach	Three layers of smooth muscle, an inner oblique, a middle circular, and an outer longitudinal; one more than the rest of the tube

The Small Intestine

The small intestine is the long, coiled region where most absorption happens. Its lining is folded again and again to expand the surface area.

THE REGIONS OF THE SMALL INTESTINE AND THE FOLDS OF ITS LINING

Structure	What it is
Duodenum	The first and shortest region, about 25 cm (10 inches), curving from the pyloric sphincter; mostly retroperitoneal
Jejunum	The middle region, about 3 feet long
Ileum	The longest region, about 6 feet, joining the large intestine at the ileocecal sphincter
Ileocecal sphincter	The valve of smooth muscle where the ileum meets the large intestine
Circular folds	The plicae circulares; permanent ridges of the mucosa and submucosa that increase surface area
Villi	Finger-like projections of the mucosa that give the lining a velvety look and greatly increase surface area
Lacteal	The lymphatic capillary inside each villus that absorbs dietary fats
Microvilli	The microscopic projections of the absorptive cells that form the brush border

KEY POINT

The small intestine amplifies its surface three times over, each fold smaller than the last: circular folds of the wall, villi of the mucosa, and microvilli on the single absorptive cell.

The Large Intestine

The large intestine is the wider, final region of the canal. It absorbs water and forms and stores the feces.

THE REGIONS AND NAMED FEATURES OF THE LARGE INTESTINE

Structure	What it is
Cecum	The pouch at the start of the large intestine, just below the ileocecal valve
Vermiform appendix	The narrow, worm-like tube of lymphatic tissue attached to the cecum
Colon	The long portion with four parts: the ascending, transverse, descending, and sigmoid colon
Rectum	The straight portion, about 6 inches, anterior to the sacrum and coccyx
Anal canal	The last 2 to 3 cm of the tract, ending at the anus
Internal and external anal sphincters	The involuntary smooth muscle and voluntary skeletal muscle that guard the anus
Teniae coli	Three thickened bands of longitudinal muscle running the length of the colon
Haustra	The pouches that give the colon its puckered appearance, formed by the teniae coli
Omental appendices	Small fat-filled pouches of peritoneum attached to the teniae coli

The Pathway of Food

Food moves through the alimentary canal in one direction, through each region in turn.

1. **Mouth** food is taken in, chewed, and mixed with saliva
2. **Pharynx** the swallowed food passes through the oropharynx and laryngopharynx
3. **Esophagus** muscular contractions carry the food down to the stomach
4. **Stomach** food is churned and held as it becomes a liquid
5. **Small intestine** the duodenum, jejunum, and ileum, where most absorption occurs
6. **Large intestine** the cecum, colon, rectum, and anal canal, where water is absorbed and feces form
7. **Anus** the feces are eliminated

Common Disorders of the Alimentary Canal

Compare the common disorders of the canal by the structure each one affects.

THE COMMON DISORDERS OF THE CANAL

Disorder	What it is
Gastroesophageal reflux disease (GERD)	Stomach contents flow back up through a weak lower esophageal sphincter and irritate the esophagus
Hiatal hernia	Part of the stomach pushes up through the esophageal hiatus of the diaphragm
Peptic ulcer	An eroded sore in the wall of the stomach or duodenum
Appendicitis	Inflammation of the vermiform appendix; rupture can spread infection to the peritoneum
Diverticulosis	Small pouches that bulge outward through weak spots in the colon wall
Inflammatory bowel disease	Chronic inflammation of the intestinal wall, as in Crohn disease and ulcerative colitis
Hemorrhoids	Swollen, dilated veins in the wall of the anal canal and rectum
Colorectal cancer	A malignant tumor of the colon or rectum

THE LOCATION OF A PROBLEM NAMES THE SYMPTOM

Because each region of the canal has its own job, where a problem sits predicts what the patient feels. A weak **lower esophageal sphincter** brings heartburn behind the breastbone; an inflamed **appendix** brings pain to the lower right abdomen.

The four-layer wall explains how a problem deepens. A **peptic ulcer** begins in the mucosa, and the danger is that it erodes through the submucosa and muscularis. If it reaches the serosa and perforates, the contents of the lumen spill into the peritoneal cavity.